

EduGenie: Google Gemini Powered Learning Assistant

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INTRODUCTION

EduGenie is a lightweight AI-powered educational assistant that simplifies learning through generative AI. Designed for students of all academic levels, EduGenie enables users to:

- Ask questions and receive smart, concise answers
- Understand complex concepts through simplified explanations
- Generate quizzes from topics or text
- Receive personalized learning recommendations
- Summarize large educational passages

User Interaction

The student first opens the EduGenie website. The website provides a simple interface where the student can select what they want to do, such as:

- Ask a question

- Explain a topic
- Generate a quiz
- Summarize text
- Get a learning path

Frontend

The frontend is the part of EduGenie that the student can see and use. It is created using **HTML and CSS**.

The frontend contains a task selection menu, a text input area, a submit button, and an area where the result is displayed.

Backend

The backend is developed using **FastAPI**, a Python framework used to handle requests from the website.

FastAPI receives the student's request and identifies which feature is required.

For example:

- A question is sent to the Question & Answer module.
- A topic explanation is sent to the Explanation module.
- A quiz request is sent to the Quiz module.
- A long passage is sent to the Summary module.
- A learning request is sent to the Learning Path module.

AI Processing

After identifying the required task, the appropriate module sends the request to the AI model.

EduGenie uses two AI models in the documented design:

Google Gemini: Used for Question & Answer, quiz generation, summarization, and learning recommendations.

Main Idea

In simple terms, EduGenie works like a **personal AI study assistant**. The student provides some information or asks for help, the system identifies what the student needs, sends the request to the appropriate AI model, and displays the result in a simple format.

This makes learning easier by providing quick answers, simple explanations, quizzes, summaries, and step-by-step learning recommendations.

Technologies Used

- Python
- Fast API-Application Programming Interface
- HTML-HyperText Markup Language
- CSS-Cascading Style Sheets
- Google Gemini API
- Uvicorn
- Jinja2
- JSON-JavaScript Object Notation

The main technologies work together as follows:

HTML + CSS → Frontend

Python + FastAPI → Backend

Gemini API + LaMini-Flan-T5 → AI Processing

Uvicorn → Runs the Application

Jinja2 → Connects Backend Data with HTML

JSON → Organizes Data

DIAGRAM



Features

- Question & Answer
- Explain
- Quiz
- Summarize
- Learning Path

Modules

- Q&A Module
- Explanation Module
- Quiz Module
- Summary Module
- Learning Path Module

7. Workflow

EduGenie follows a simple process to provide AI-based learning support to the user.

Step 1: User Input

The student opens the EduGenie website and selects the required task, such as **Ask, Explain, Quiz, Summarize, or Learning Path**. The student then enters the required question, topic, or text.

Step 2: Request to Backend

The student's input is sent from the website to the **FastAPI backend**. The backend receives the request and identifies which feature the student has selected.

Step 3: Select Module

FastAPI sends the request to the appropriate module.

- **Q&A** → Question Answering Module
- **Explain** → Explanation Module
- **Quiz** → Quiz Module
- **Summarize** → Summary Module
- **Learn** → Learning Path Module

Step 4: AI Processing

The selected module sends the request to the appropriate AI model.

Gemini is used for Q&A, quiz generation, summarization, and learning recommendations.

LaMini-Flan-T5 is used for simple concept explanations.

Step 5: Generate Result

The AI processes the student's input and generates the required response. For quizzes, the system generates three multiple-choice questions with four options each.

Step 6: Display Result

The generated response is sent back through the FastAPI backend to the website. The student can then view the answer, explanation, quiz, summary, or learning recommendations.

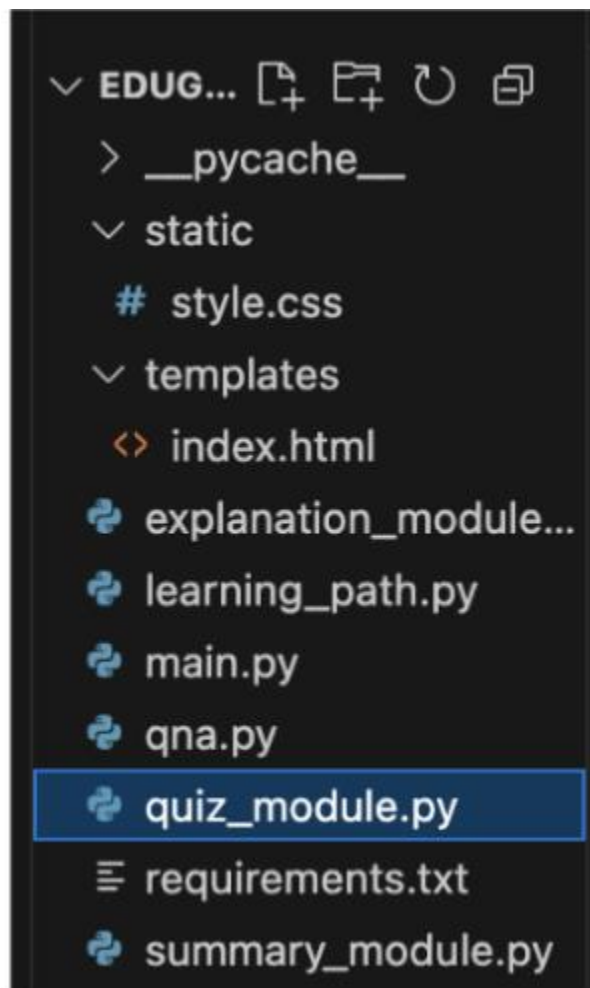
Simple Workflow

Student Input → Frontend → FastAPI → Selected Module → AI Model → Result → Frontend → Student

This workflow allows EduGenie to provide different learning services through one simple interface. The documented API endpoints are /qa, /explain, /quiz, /summarize, and /learn/recommendations.

MILESTONE 1: Model Selection and Architecture

FIG. Showing the Folder Architect



MILESTONE 2: Core Functionalities Development

EduGenie uses the **LaMini-Flan-T5** AI model to explain difficult topics in simple and easy-to-understand language. It provides short and clear explanations, making it useful for beginners, school students, and self-learners.

```
planation_module.py
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
import torch

# Load the improved model for explanations
explain_tokenizer = AutoTokenizer.from_pretrained("MBZUAI/LaMini-Flan-T5-783M")
explain_model = AutoModelForSeq2SeqLM.from_pretrained("MBZUAI/LaMini-Flan-T5-783M")

def explain_topic(topic: str) -> str:
    input_text = f"Explain the concept of '{topic}' in a simple and clear way for a school student."
    inputs = explain_tokenizer(input_text, return_tensors="pt")

    outputs = explain_model.generate(
        **inputs,
        max_new_tokens=150,
        temperature=0.7,
        top_k=50,
        top_p=0.95,
        do_sample=True
    )

    explanation = explain_tokenizer.decode(outputs[0], skip_special_tokens=True)
    return explanation
```

Fig.Explanation module logic

Q&A Module

EduGenie uses **Gemini 1.5 Pro** to answer general knowledge and academic questions. It understands the user's question and provides clear and useful answers on different educational topics.

```
def answer_question_with_gemini(question: str) -> str:
    try:
        model = genai.GenerativeModel(model_name="models/gemini-1.5-pro")
        response = model.generate_content(question)
        return response.text.strip()
    except Exception as e:
        return f"⚠ Error in QnA: {e}"
```

Quiz Module

EduGenie uses **Gemini** to create **3 multiple-choice questions** from a given passage. Each question has **4 options** and one correct answer. The questions are returned in **JSON format** so the system can easily display them.

Summary Module

This feature leverages Gemini's generative capabilities to summarize long paragraphs into concise, easy-to-understand versions, making it ideal for quick revision. It ensures the core information is retained while eliminating redundancy, helping learners absorb key points efficiently without losing clarity or context.

Learning Path Module

EduGenie uses **Google Gemini** to create a personalized learning path for any topic. It provides topics from **beginner to advanced levels** along with useful resources such as videos, articles, and books. It also handles errors and provides a helpful message if the response is incomplete or invalid

MILESTONE 3: Frontend Developmen

Build Web Interface

HTML Form (index.html):

- Task dropdown (Explain, QnA, Quiz, Summary, Recommend Path)
- Text area for user input
- Submit button

CSS Styling (style.css):

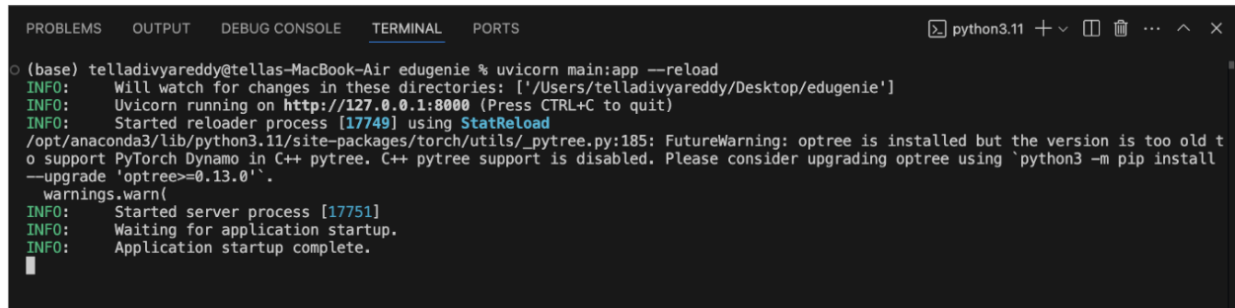
- Responsive design
- Styled buttons, input areas, and result container

Live Integration

- Form submission sends POST request to FastAPI backend
- Result appears below input box in real-time

MILESTONE 4: Deployment

- Run Locally
Run locally with `uvicorn main:app --reload`
Navigate to `http://127.0.0.1:8001/`



```
(base) telladivvyareddy@tellas-MacBook-Air edugenie % uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['/Users/telladivvyareddy/Desktop/edugenie']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [17749] using StatReload
/opt/anaconda3/lib/python3.11/site-packages/torch/utils/_pytree.py:185: FutureWarning: optree is installed but the version is too old to support PyTorch Dynamo in C++ pytree. C++ pytree support is disabled. Please consider upgrading optree using `python3 -m pip install --upgrade 'optree>=0.13.0'`.
warnings.warn(
INFO: Started server process [17751]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

- Functional Testing
Ask a question
Get an explanation
Generate a quiz
Summarize content
Get a personalized learning plan

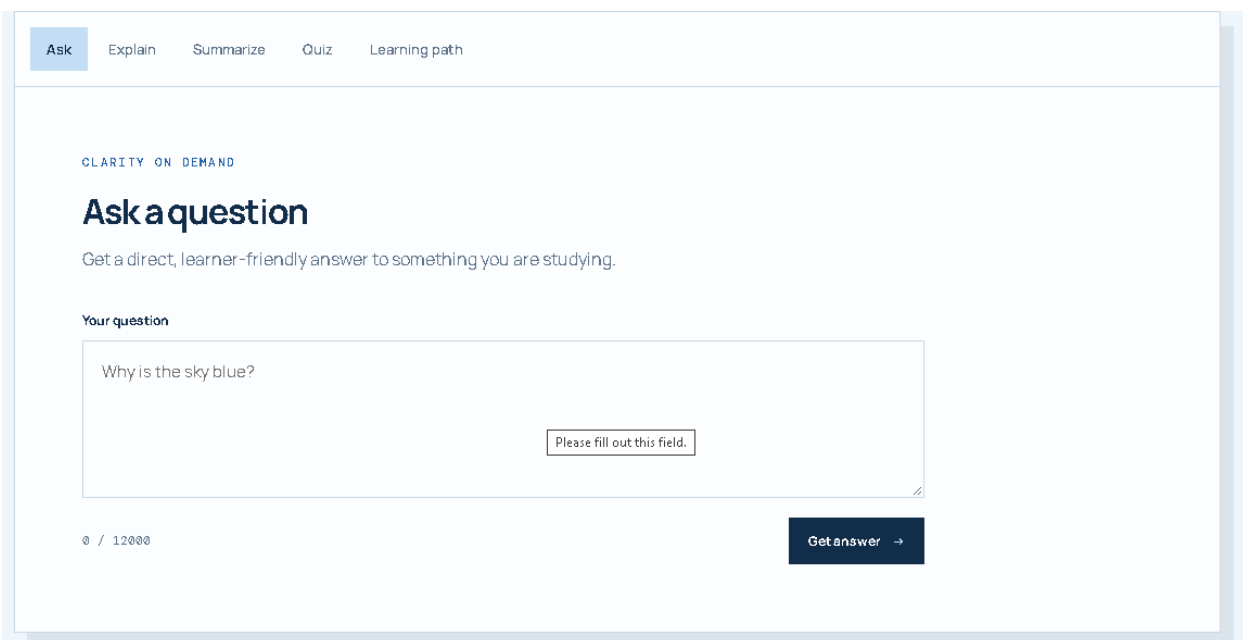


Fig. EDUGENIE

Ask

Explain

Summarize

Quiz

Learning path

MAKE IT CLICK

Explain a topic

Break down a difficult concept with a plain-language explanation.

Topic

Photosynthesis

0 / 12000

Explain topic →

Exploring EduGenie

Ask

Explain

Summarize

Quiz

Learning path

FIND THE SIGNAL

Summarize notes

Reduce a long passage to the ideas worth remembering.

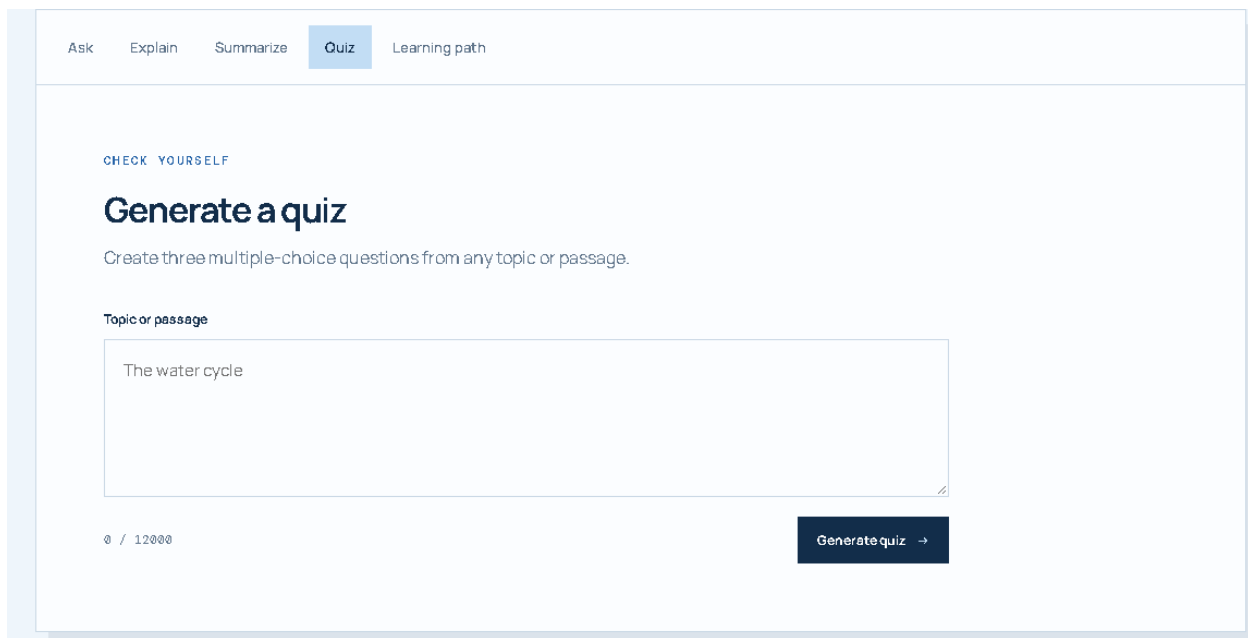
Passage

Paste your study notes here...

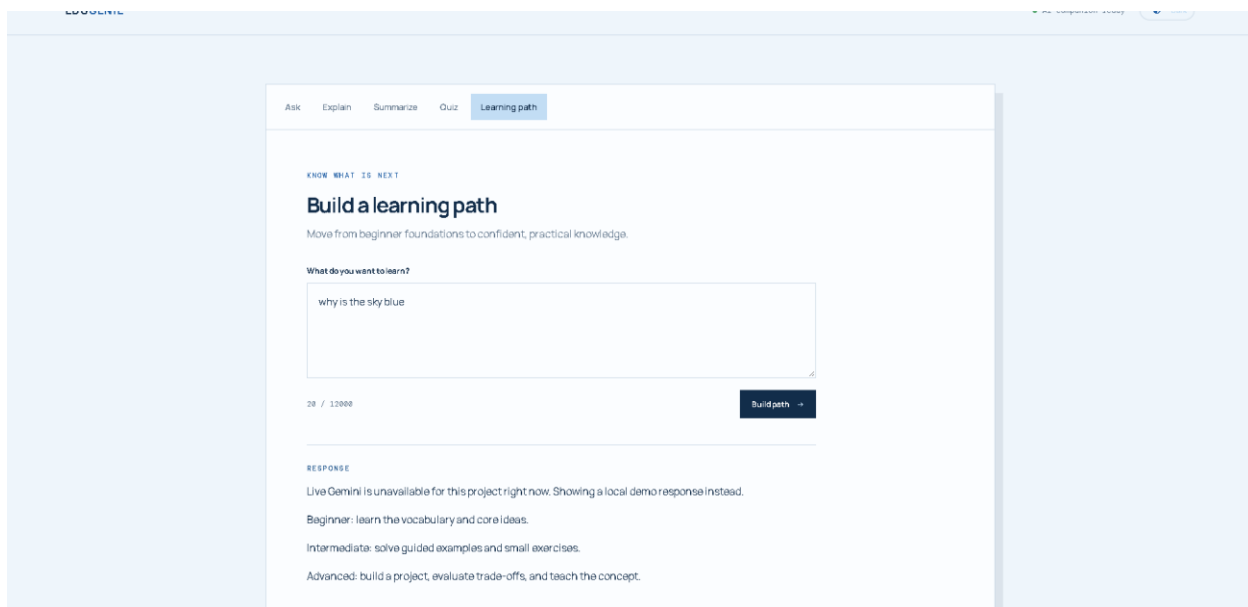
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Summarize →

Summarize note



QUIZ



LEARNING PATH

Conclusion

EduGenie demonstrates how Generative AI can be effectively integrated into an educational platform to provide personalized, interactive, and accessible

learning support. The project combines a modern web-based interface with AI-powered capabilities such as question answering, topic explanation, summarization, quiz generation, answer evaluation, and personalized learning-path recommendations.

By integrating the Gemini API with a structured backend and responsive frontend, EduGenie provides students with an intelligent learning assistant that can adapt its responses to different topics and learning requirements. The use of structured AI outputs and validation mechanisms also improves the reliability and consistency of generated educational content.

The project successfully demonstrates the practical application of Generative AI in education while maintaining a modular architecture that can be extended with additional features in the future. Possible enhancements include user authentication, progress tracking, personalized student profiles, multilingual learning support, voice-based interaction, richer educational resources, and integration with school or learning-management systems.

Overall, EduGenie shows the potential of Generative AI to complement traditional education by making learning more personalized, interactive, and readily accessible to students.
